Chapter 3. Continuous Functions

Limits and continuity, extreme and intermediate values, uniform continuity, monotone functions

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Throughout, $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$, and $f : X \to Y$.

3.1 Limits and continuity

Definition (Limit and continuity)

Let $x_0 \in X'$. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ such that $x \in X$, $0 < \|x - x_0\| < \delta$ imply $\|f(x) - y_0\| < \epsilon$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$. It is continuous on X if continuous at every point.

At an isolated point every function is continuous; at a limit point $x_0 \in X$, continuity means $\lim_{x\to x_0} f(x) = f(x_0)$.

Proposition 3.1 (Sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X. Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for all $x_k \to x_0$ with $x_k \neq x_0$.

Proof. Necessity is immediate from the definitions. If f is not continuous at x_0, there is $\epsilon > 0$ such that for each k some x_k with $\|x_k - x_0\| < 1/k$ has $\|f(x_k) - f(x_0)\| \geq \epsilon$; this sequence violates the criterion. ∎

Consequently sums, scalar multiples, products and quotients (where defined, nonzero denominator) of continuous functions are continuous, and compositions of continuous functions are continuous. A map into $\mathbb{R}^m$ is continuous iff each component is. Polynomials in n variables are continuous.

Theorem 3.2 (Topological characterization)

The following are equivalent: (a) $f : X \to Y$ is continuous; (b) $f^{-1}(V)$ is open in X for every open $V \subset \mathbb{R}^m$; (c) $f^{-1}(F)$ is closed in X for every closed $F \subset \mathbb{R}^m$.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$, choose ϵ with $N(f(x), \epsilon) \subset V$ and the corresponding δ; then $N(x, \delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): apply (b) to $V = N(f(x_0), \epsilon)$. (b)⇔(c) by taking complements. ∎

Example 3.1

(a) $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$, is continuous in each variable separately but not at the origin: along $y = mx$ it equals $m/(1 + m^2)$.
(b) The function equal to $1/q$ at $x = p/q$ in lowest terms ($q \geq 1$) and 0 at irrationals is continuous exactly at the irrationals: near any x, only finitely many rationals have denominator $\leq q$.
(c) $\{x : f(x) > 0\}$ is open for continuous real f on $\mathbb{R}^n$; $\{x : f(x) = 0\}$ is closed.

3.2 Compactness, connectedness and continuity

Theorem 3.3

The continuous image of a compact set is compact.

Proof. If $\{V_i\}$ is an open cover of $f(X)$, the sets $f^{-1}(V_i)$ are relatively open and cover X: $f^{-1}(V_i) = X \cap U_i$ with U_i open. Finitely many U_i cover X, so finitely many V_i cover $f(X)$. (Alternatively, use sequential compactness.) ■

Theorem 3.4 (Extreme value theorem)

A continuous real function on a nonempty compact set attains a maximum and a minimum. In particular, if $f > 0$ on compact X, then $f \geq c$ for some constant $c > 0$.

Proof. $f(X)$ is compact, hence closed and bounded, so it contains $\sup f(X)$ and $\inf f(X)$. ■

Proposition 3.5

If $f : X \to Y$ is a continuous bijection and X is compact, then f^{-1} is continuous.

Proof. For $F \subset X$ closed, F is compact, so $(f^{-1})^{-1}(F) = f(F)$ is compact, hence closed. Apply Theorem 3.2(c). ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection from $[0, 2\pi)$ onto the circle whose inverse is discontinuous at $(1, 0)$.

Proposition 3.6

If $A \subset \mathbb{R}^n$ is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\} > 0$.

Proof. $x \mapsto \operatorname{dist}(x, B)$ is continuous (indeed $|\operatorname{dist}(x, B) - \operatorname{dist}(y, B)| \leq \|x - y\|$) and positive on A since B is closed; apply Theorem 3.4. (Both sets merely closed is not enough: consider the x-axis and the graph of $1/x$.) ■

Theorem 3.7

The continuous image of a connected set is connected.

Proof. If open U, V separated $f(X)$, then $f^{-1}(U), f^{-1}(V)$ would separate X. ■

Corollary 3.8 (Intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \gamma < f(b)$, then $f(c) = \gamma$ for some $c \in (a, b)$. The continuous image of an interval is an interval, and the image of $[a, b]$ is a closed bounded interval.

Proof. $f([a, b])$ is connected, hence an interval (Proposition 2.14); combined with Theorem 3.3 it is $[\min f, \max f]$. ■

Example 3.2

(a) Every continuous $f : [a, b] \to [a, b]$ has a fixed point: apply the intermediate value theorem to $f(x) - x$.

(b) Every polynomial of odd degree has a real root. Every $y > 0$ has a unique positive n-th root.
(c) No continuous $f : \mathbb{R} \to \mathbb{R}$ takes every value exactly twice. (Consider the maximum on the interval between the two preimages of some value.)

3.3 Uniform continuity

Definition (Uniform continuity)

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ such that $x, x' \in X$, $\|x - x'\| < \delta$ imply $\|f(x) - f(x')\| < \epsilon$.

The difference from continuity is that δ may not depend on the point. $x \mapsto 1/x$ on $(0, 1)$ and $x \mapsto x^2$ on $\mathbb{R}$ are continuous but not uniformly continuous; Lipschitz functions ($\|f(x) - f(x')\| \le L\|x - x'\|$) are uniformly continuous.

Theorem 3.9 (Heine)

A continuous function on a compact set is uniformly continuous.

Proof. If not, there are $\epsilon > 0$ and x_k, x'_k with $\|x_k - x'_k\| < 1/k$ but $\|f(x_k) - f(x'_k)\| \ge \epsilon$. By compactness a subsequence $x_{k_j} \to x \in X$, and then $x'_{k_j} \to x$ too; continuity at x gives $f(x_{k_j}) - f(x'_{k_j}) \to 0$, a contradiction. ■

Continuous maps preserve convergent sequences but not Cauchy sequences ($1/x$ maps the Cauchy sequence $1/k$ in $(0, 1)$ to an unbounded one). Uniformly continuous maps do preserve them:

Theorem 3.10

If $f : X \to \mathbb{R}^m$ is uniformly continuous, it maps Cauchy sequences to Cauchy sequences and extends uniquely to a continuous (indeed uniformly continuous) function on $\overline{X}$.

Proof. The first claim is immediate. For $x \in \overline{X}$ choose $x_k \to x$ in X; $(f(x_k))$ is Cauchy, hence convergent, and the limit does not depend on the sequence (interleave two sequences). Define $\tilde{f}(x)$ as this limit; the ϵ–δ estimate passes to the limit. ■

So a continuous function on a bounded set X is uniformly continuous iff it extends continuously to the compact set $\overline{X}$.

3.4 Monotone functions

Theorem 3.11

A monotone function f on an open interval (a, b) has one-sided limits $f(x-)$ and $f(x+)$ at every point. If f is increasing, $f(x-) \le f(x) \le f(x+)$, and $f(x+) \le f(y-)$ for $x < y$.

Proof. For increasing f, $f(x-) = \sup\{f(t) : t < x\}$: given ϵ, some $t_0 < x$ has $f(t_0) > f(x-) - \epsilon$, and monotonicity gives the estimate for all $t \in (t_0, x)$. Similarly $f(x+) = \inf_{t>x} f(t)$. ■

Corollary 3.12

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, at each discontinuity x choose a rational $r(x) \in (f(x-), f(x+))$. These open intervals are disjoint for different x (by the last inequality of Theorem 3.11), so $x \mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{q_k\}$ the increasing function $\sum_k 2^{-k}\mathbf{1}\{x > q_k\}$ is discontinuous exactly at the q_k.

Theorem 3.13

For a continuous f on an interval I the following are equivalent: (a) f is injective; (b) f is strictly monotone. In that case $f(I)$ is an interval and $f^{-1} : f(I) \to I$ is continuous and strictly monotone.

Proof. (a)⇒(b): if f were not monotone, there would be $x < y < z$ with $f(y)$ not between $f(x)$ and $f(z)$, and the intermediate value theorem would produce a repeated value. For the inverse: f^{-1} is strictly monotone on the interval $f(I)$, and its image I is an interval, so it has no jumps (a jump would leave a gap in the image), i.e. it is continuous. ■

This gives continuity of $x^{1/n}$, $\log x$, $\arcsin x$, etc., from that of their inverses.